Irreducibility, Prime Ideals, and Coordinate Domains

An End-to-End Meta-Math Walkthrough

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Abstract

A product vanishes at each point exactly when at least one factor does. This translates a geometric split into zero divisors in the coordinate ring. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

AG-04 / Algebraic Geometry / Report v1

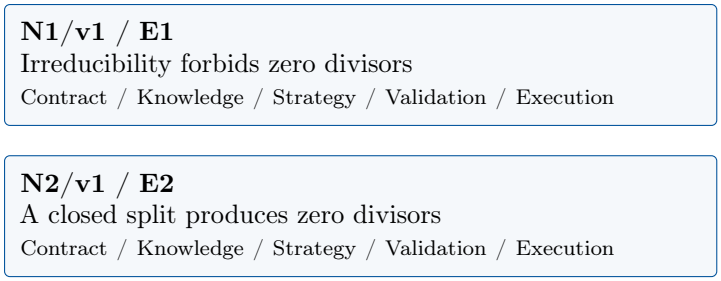
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let $X \subset k^n$ be a nonempty affine algebraic set over a field k. With its Zariski subspace topology, X is irreducible if and only if $I(X)$ is prime, equivalently the coordinate ring $k[X] = k[x_1, \ldots, x_n]/I(X)$ is an integral domain. Irreducible means not a union of two proper closed subsets. Algebraic closedness is not needed for this statement about the actual point set and its vanishing ideal.*

A. Foundations and permitted imports

Use $V(fg) = V(f) \cup V(g)$ over a field, the definition of the Zariski topology, and the quotient criterion: a proper ideal P is prime iff $ab \in P$ implies $a \in P$ or $b \in P$. The quotient is a domain precisely when this implication holds. Relative closed sets in X are $X \cap V(S)$ for collections S of polynomials.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

No inter-node proof imports are needed: the nodes establish separate directions under the common theorem contract.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Irreducibility forbids zero divisors

Object and role. Convert a vanishing product into a two-set closed cover. **Local obligation:** If X is irreducible then $I(X)$ is prime.

Inputs. Nonempty irreducible X and $fg \in I(X)$.

C — Contract

Publish E1/v1 only after establishing this obligation: If X is irreducible then $I(X)$ is prime.

K — Knowledge

Nonempty irreducible X and $fg \in I(X)$. Retain the stated assumptions and import versions.

S — Strategy

Convert a vanishing product into a two-set closed cover.

V — Validation

Nonemptiness makes $I(X)$ proper; the empty set has the unit vanishing ideal.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 Nonemptiness gives $1 \notin I(X)$. If $fg \in I(X)$, each point of X lies in $V(f)$ or $V(g)$, because a field has no zero divisors. Hence $X = (X \cap V(f)) \cup (X \cap V(g))$.

N1.2 Both pieces are closed in X. Irreducibility forces one to equal X, which says $f \in I(X)$ or $g \in I(X)$. This proves primality.

Export E1/v1. If X is irreducible then $I(X)$ is prime.

Acceptance gate. Nonemptiness makes $I(X)$ proper; the empty set has the unit vanishing ideal.

Targeted repair. If the empty set is included as irreducible, restore nonemptiness before claiming a proper prime ideal. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: A closed split produces zero divisors

Object and role. Choose one polynomial detecting a point outside each closed subset and multiply them. **Local obligation:** If $I(X)$ is prime then X is irreducible, and both conditions are equivalent to $k[X]$ being a domain.

Inputs. Prime $I(X)$ and a proposed decomposition $X = Y \cup Z$ into two proper relatively closed subsets.

C — Contract

Publish E2/v1 only after establishing this obligation: If $I(X)$ is prime then X is irreducible, and both conditions are equivalent to $k[X]$ being a domain.

K — Knowledge

Prime $I(X)$ and a proposed decomposition $X = Y \cup Z$ into two proper relatively closed subsets. Retain the stated assumptions and import versions.

S — Strategy

Choose one polynomial detecting a point outside each closed subset and multiply them.

V — Validation

A proper closed subset has a defining equation that fails somewhere on X; do not assume one equation defines the whole subset.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 Write $Y = X \cap V(S)$ and choose $a \in X \setminus Y$. Some $f \in S$ has $f(a) \neq 0$, so f vanishes on Y but is not in $I(X)$. Similarly choose g vanishing on Z but not on all of X.

N2.2 The product fg vanishes on $Y \cup Z = X$, so $fg \in I(X)$ while neither factor belongs to it, contradicting primality. Thus a proper split is impossible. Finally, multiplication of residue classes in $k[X]$ is zero exactly when the product of representatives lies in $I(X)$; the prime-ideal condition is exactly absence of nonzero zero divisors, with $1 \neq 0$.

Export E2/v1. If $I(X)$ is prime then X is irreducible, and both conditions are equivalent to $k[X]$ being a domain.

Acceptance gate. A proper closed subset has a defining equation that fails somewhere on X; do not assume one equation defines the whole subset.

Targeted repair. If a single defining equation for Y is assumed without justification, select just one member of its defining family that fails at a. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Irreducibility forbids zero divisors

1. **N1.1.** Nonemptiness gives $1 \notin I(X)$. If $fg \in I(X)$, each point of X lies in $V(f)$ or $V(g)$, because a field has no zero divisors. Hence $X = (X \cap V(f)) \cup (X \cap V(g))$.

2. **N1.2.** Both pieces are closed in X. Irreducibility forces one to equal X, which says $f \in I(X)$ or $g \in I(X)$. This proves primality.

B. N2 — A closed split produces zero divisors

1. **N2.1.** Write $Y = X \cap V(S)$ and choose $a \in X \setminus Y$. Some $f \in S$ has $f(a) \neq 0$, so f vanishes on Y but is not in $I(X)$. Similarly choose g vanishing on Z but not on all of X.

2. **N2.2.** The product fg vanishes on $Y \cup Z = X$, so $fg \in I(X)$ while neither factor belongs to it, contradicting primality. Thus a proper split is impossible. Finally, multiplication of residue classes in $k[X]$ is zero exactly when the product of representatives lies in $I(X)$; the prime-ideal condition is exactly absence of nonzero zero divisors, with $1 \neq 0$.

□

V. WORKED ORCHESTRATION

In $\mathbb{C}^2$, the set $X = V(xy)$ is the union of the two coordinate axes. The classes of x and y are nonzero functions on X, but their product is zero. For the parabola $Y = V(y - x^2)$, evaluation $x \mapsto t$, $y \mapsto t^2$ identifies its coordinate ring with $\mathbb{C}[t]$: division by the monic polynomial $y - x^2$ leaves a remainder in x, which vanishes at every complex x only if zero. Thus the parabola is irreducible.

A. Boundary and hypothesis audit

Connectedness is weaker than irreducibility. The two complex coordinate axes meet at the origin, so their union is connected in the usual topology, yet the two proper algebraic axes split it. Algebraic irreducibility is the condition proved here.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N2 inference: every proper closed subset is a hypersurface. For example the origin in $\mathbb{C}^2$ requires the simultaneous equations $x = y = 0$ as a set of two independent coordinate restrictions. **Repair:** select one defining polynomial nonzero at a chosen outside point; the argument needs a separator, not a single equation for the entire subset. This corrects the proof mechanism without changing the target. Recheck the product witness and domain conclusion.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Use domain coordinate rings to define function fields of irreducible affine varieties.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. J. S. Milne, Algebraic Geometry, chapters on commutative algebra, algebraic sets, and affine varieties; author's mathematics site.